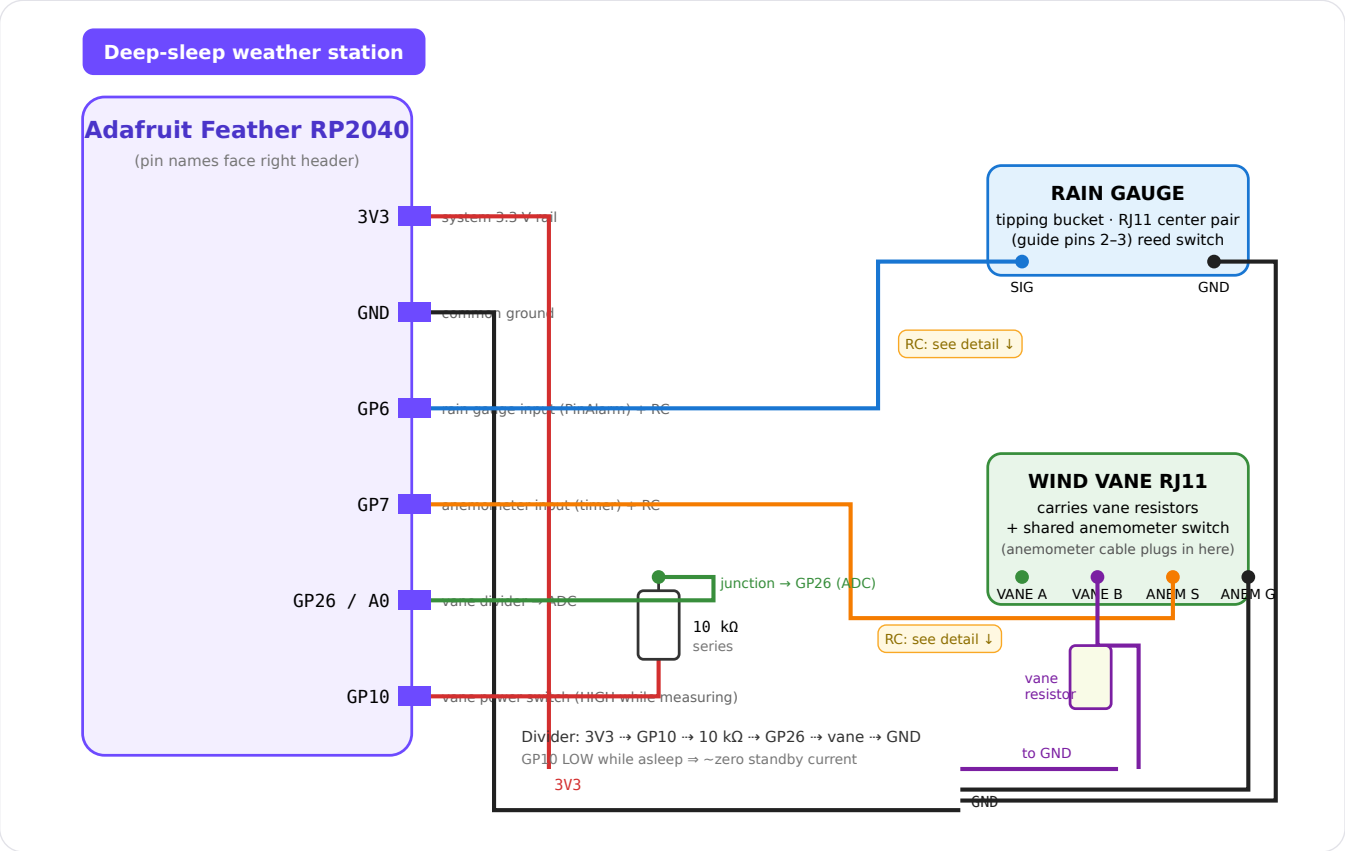
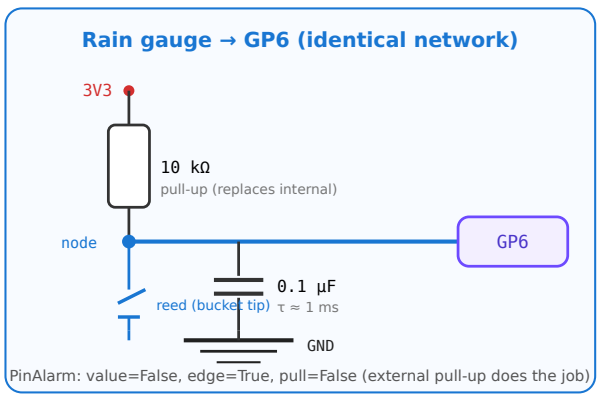
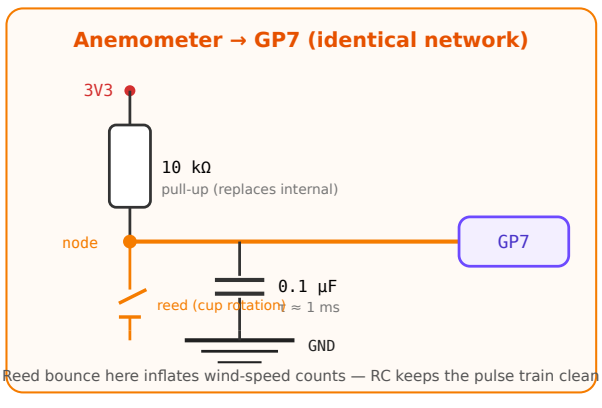
Feather RP2040 ↔ SparkFun Weather Meter Kit (SEN-15901)

Deep-sleep weather station wiring · rain gauge wakes the chip (PinAlarm), anemometer sampled on timer, vane read by ADC — now with RC debounce networks on both reed-switch inputs



Input conditioning detail — RC debounce networks





3.3 V GND Rain gauge signal → GP6 Anemometer signal → GP7

Vane divider node → GP26 Vane resistor leg

Notes on the debounce networks

- **Why RC instead of code-only:** while the Feather is in deep sleep there is *no software running* — a bouncing reed contact during a PinAlarm wake fires several spurious boots/rain-counts before any debounce code executes. The RC network is the only thing filtering sleepy pulses.
- **The 10 kΩ external pull-up replaces the internal one**: set `pull=False` in the PinAlarm (or leave the internal pull-up on and drop the external resistor — either works; the external one gives a stiffer, better-defined rail and survives if you later use the pin before configuring it).
- **τ = R·C = 10 kΩ × 0.1 µF ≈ 1 ms.** Reed contacts in this kit settle in well under 1 ms, while bounce lasts a few ms — the cap absorbs the chatter but passes every genuine tip/rotation. If you see missed pulses in strong winds (fast anemometer spinning), shrink C to 0.047 µF; if you still see double-counts on rain, raise it to 0.47 µF (but don't exceed ~10 ms or very fast cup rotation starts to smear).
- **Double protection:** keep the software debounce too (ignore anemometer pulses < 5 ms apart while awake) — the RC handles asleep-time, the code handles awake-time.
- **Optional upgrade:** for a truly clean edge on the rain line, buffer the node through one gate of a 74HC14 Schmitt-trigger inverter (powered from 3V3) and feed the output to GP6; that gives a crisp monotonic transition regardless of contact condition. Overkill for most installs, but cheap insurance for the rain counter, where every pulse is money.
- Keep the cap physically close to the Feather's header pins, and keep reed leads twisted pairs (signal+GND) on long runs to avoid picking up wind-blown static noise.